

Lava Architecture

This document collects two architectural surfaces that span multiple modules:

1. **Multi-Tracker SDK** — the layered, pluggable tracker stack added in SP-3a (1.2.0). Pure-Kotlin core, Hilt-injected orchestrator, per-tracker plugin modules, generic primitives in a vasic-digital submodule.
2. **Local Network Discovery** — the mDNS / NsdManager flow that auto-discovers the lava-api-go service on the LAN.

For per-feature MVI shapes, see the relevant feature/*/README.md (or the source — every feature mirrors XxxState + XxxAction + XxxSideEffect + XxxViewModel).

Multi-Tracker SDK (SP-3a, 1.2.0)

Layering

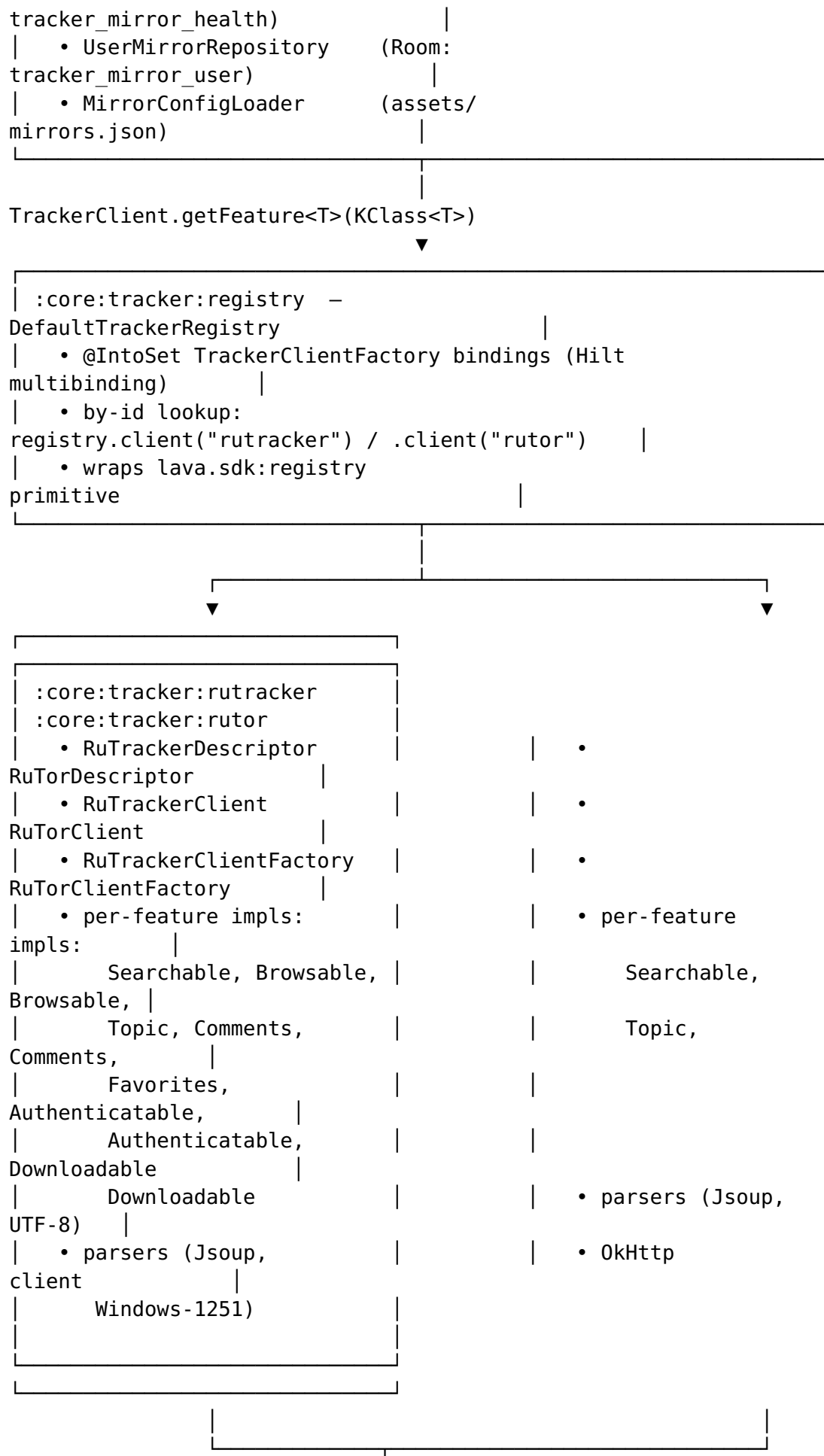
```
| feature:* ViewModels (search_result, topic, login,  
| provider_config, |  
|   credentials_manager, menu) – no per-tracker imports; talk  
to   |  
|   LavaTrackerSdk  
only
```

LavaTrackerSdk

| inject @Singleton



```
| :core:tracker:client – LavaTrackerSdk (Hilt  
orchestrator) |  
| • activeTracker():  
TrackerClient |  
| • search(req) / browse(...) / topic(...) /  
download(...) |  
| • CrossTrackerFallbackPolicy →  
CrossTrackerFallbackProposed |  
| • MirrorHealthCheckWorker (15-min  
PeriodicWork) |  
| • MirrorHealthRepository (Room:
```





```
| :core:tracker:api      - feature interfaces + capability enum +  
DTOs |  
|   • Searchable, Browsable, Topic, Comments,  
Favorites, |  
|   Authenticatable,  
Downloadable |  
|   • TrackerCapability {SEARCH, BROWSE, FORUM, TOPIC,  
COMMENTS, |  
|   FAVORITES, TORRENT_DOWNLOAD, MAGNET_LINK,  
AUTH_REQUIRED, |  
|   CAPTCHA_LOGIN, RSS, UPLOAD,  
USER_PROFILE} |  
|   • TorrentItem, SearchRequest, SearchResult, TopicDetail,  
... |  
|   • AuthType {ANONYMOUS, FORM_LOGIN,  
CAPTCHA_LOGIN} |
```

| depends-on



```
| :core:tracker:mirror  typealias →  
lava.sdk:mirror.MirrorConfigStore |  
| :core:tracker:registry      → lava.sdk:registry  
primitives |  
| :core:tracker:testing  FakeTrackerClient + builders +  
fixtures |  
|                                     ↳ lava.sdk:testing  
primitives |
```

| composite-build pin (frozen by

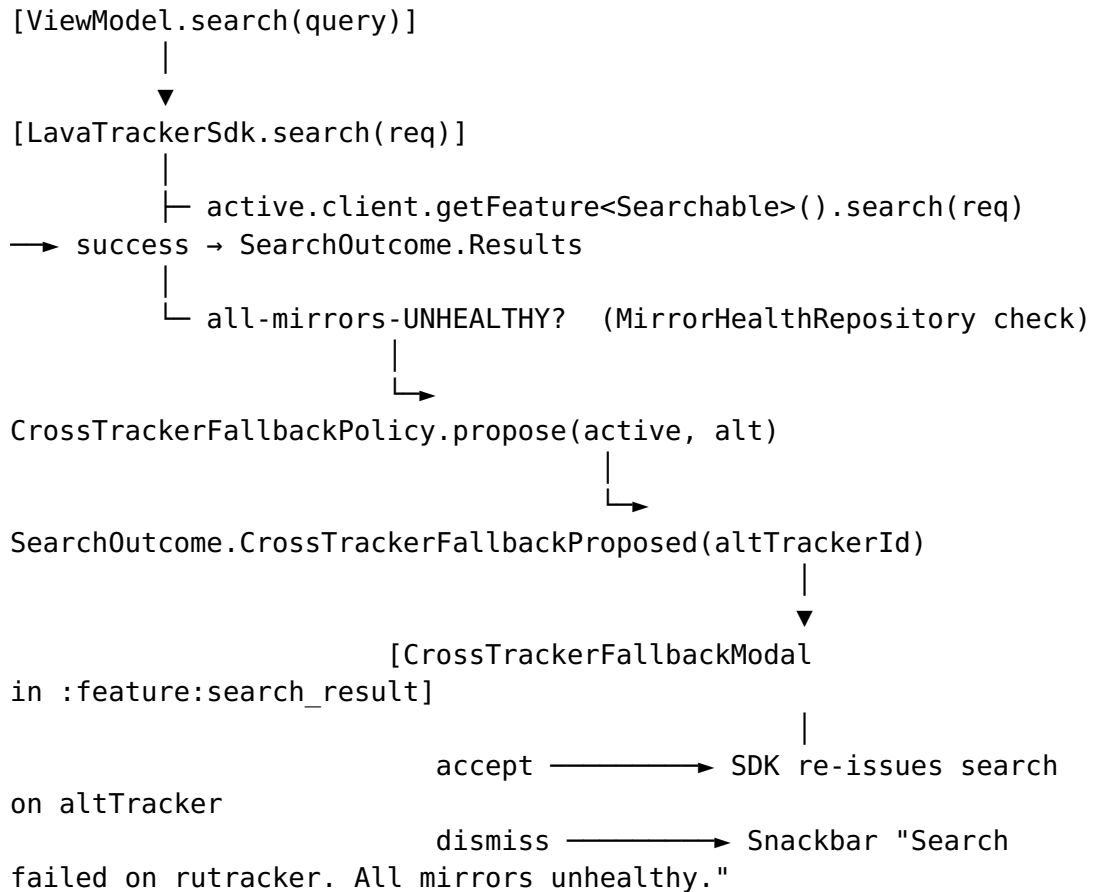
default)



```
| submodules/tracker_sdk/      (vasic-digital/Tracker-  
SDK) |  
|   • api      - MirrorUrl, Protocol, Mirror state  
machine |  
|   • mirror   - MirrorConfigStore  
interface |  
|   • registry - generic in-memory tracker registry  
primitive |  
|   • testing  - clock, dispatcher, fixture  
loader |  
|  
|  
|   Mirrored to GitHub + GitLab. Pin updates are deliberate  
PRs. |
```

The capability enum is the single source of truth. Constitutional clause 6.E (Capability Honesty) requires that **every capability declared in a TrackerDescriptor resolves to a non-null TrackerClient.getFeature<T>()**; the `:core:tracker:registry` test enumerates descriptors + capabilities and asserts non-null at JVM-test time.

Cross-tracker fallback flow



There is **no silent fallback**. The user is always shown the modal and must explicitly accept or dismiss.

Persistence

Room database `AppDatabase v7` owns two SP-3a tables (migration `v6 → v7`):

Table	Entity	Purpose
tracker_mirror_health	MirrorHealthEntity	Per-mirror probe results (HEALTHY/DEGRADED/UNHEALTHY)
tracker_mirror_user	UserMirrorEntity	User-added custom mirrors per tracker

Bundled defaults live at core/tracker/client/src/main/assets/mirrors.json and are merged at app start by MirrorConfigLoader.

Module reference

Path	Plugin	Purpose
core/tracker/api	lava.kotlin.library + serialization	Public feature interfaces + capability enum + DTOs
core/tracker/registry	lava.kotlin.library	Lava-domain TrackerRegistry over lava.sdk:registry
core/tracker/mirror	lava.kotlin.library + serialization	MirrorConfigStore typealias bridge
core/tracker/client	lava.android.library + Hilt + serialization	LavaTrackerSdk, persistence repos, WorkManager
core/tracker/rutacker	lava.kotlin.tracker.module	RuTracker plugin
core/tracker/rutor	lava.kotlin.tracker.module	RuTor plugin
core/tracker/testing	lava.kotlin.library	FakeTrackerClient + builders + fixture loaders
feature/provider_config	lava.android.feature + Compose	Per-provider configuration (mirrors, credentials binding, sync toggle, anonymous mode, clone). Reachable from Menu by tapping a provider row. Replaces the SP-3a Trackers screen (deleted in SP-4 Phase C).
feature/credentials_manager	lava.android.feature + Compose	Passphrase-gated credentials CRUD UI (Phase A). Add / edit / delete

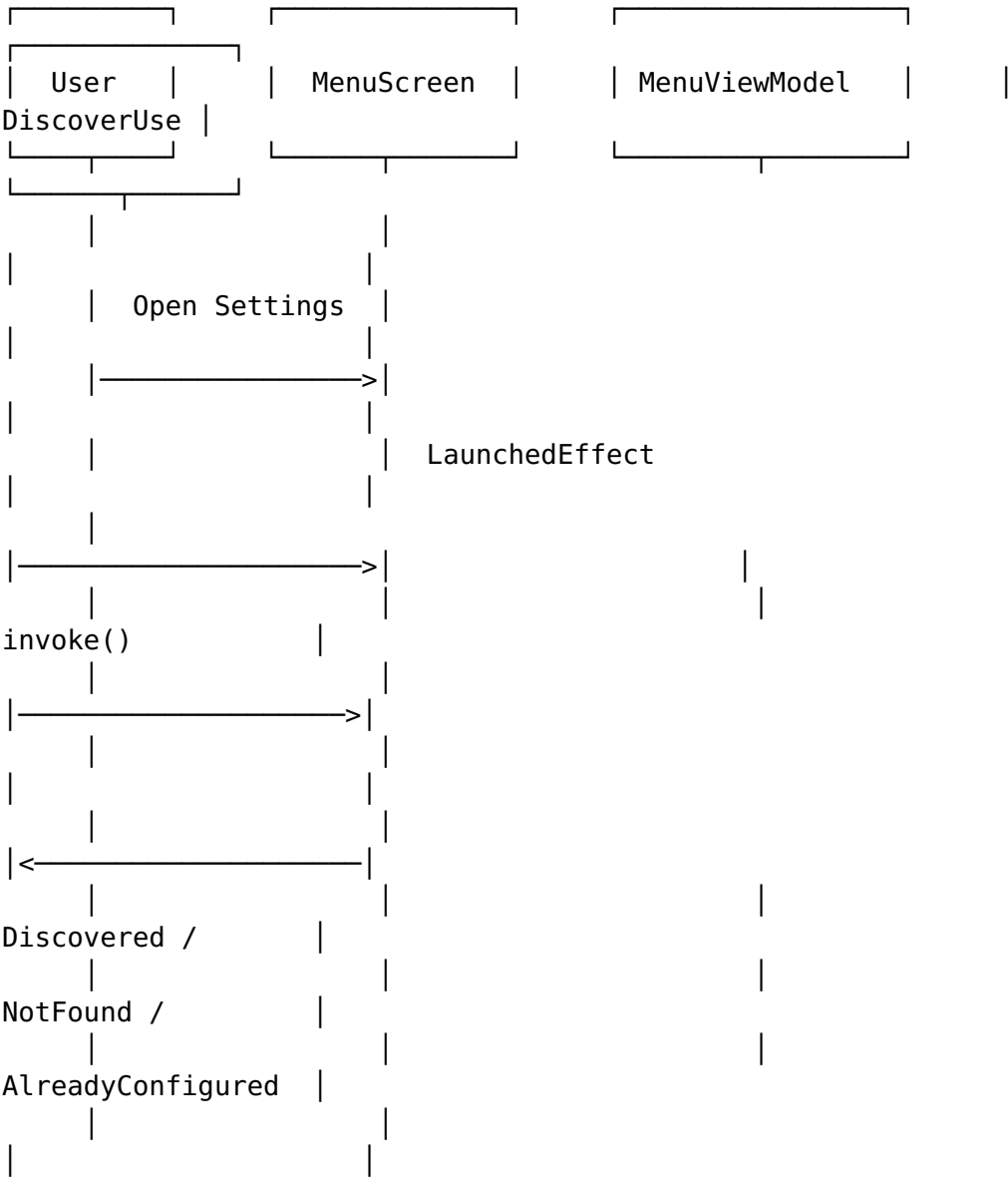
Path	Plugin	Purpose
		CredentialsEntry rows.

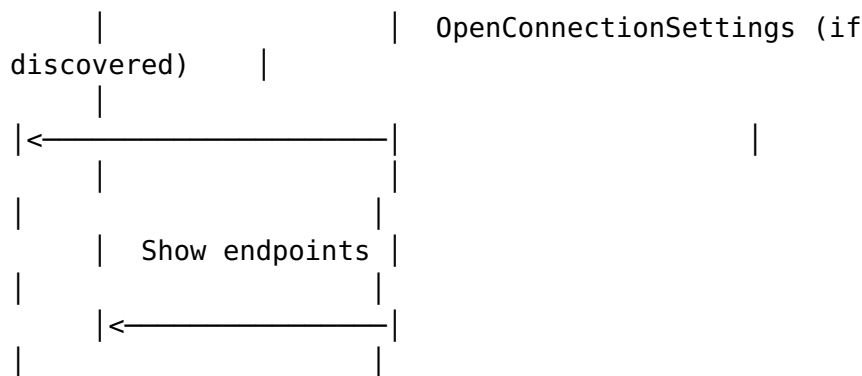
Generic primitives mounted via composite build at submodules/tracker_sdk/ (submodule pin frozen by default per the Decoupled Reusable Architecture constitutional rule).

For the full design rationale see [docs/superpowers/specs/2026-04-30-sp3a-multi-tracker-sdk-foundation-design.md](#). For the step-by-step recipe to add a third tracker see [docs/sdk-developer-guide.md](#).

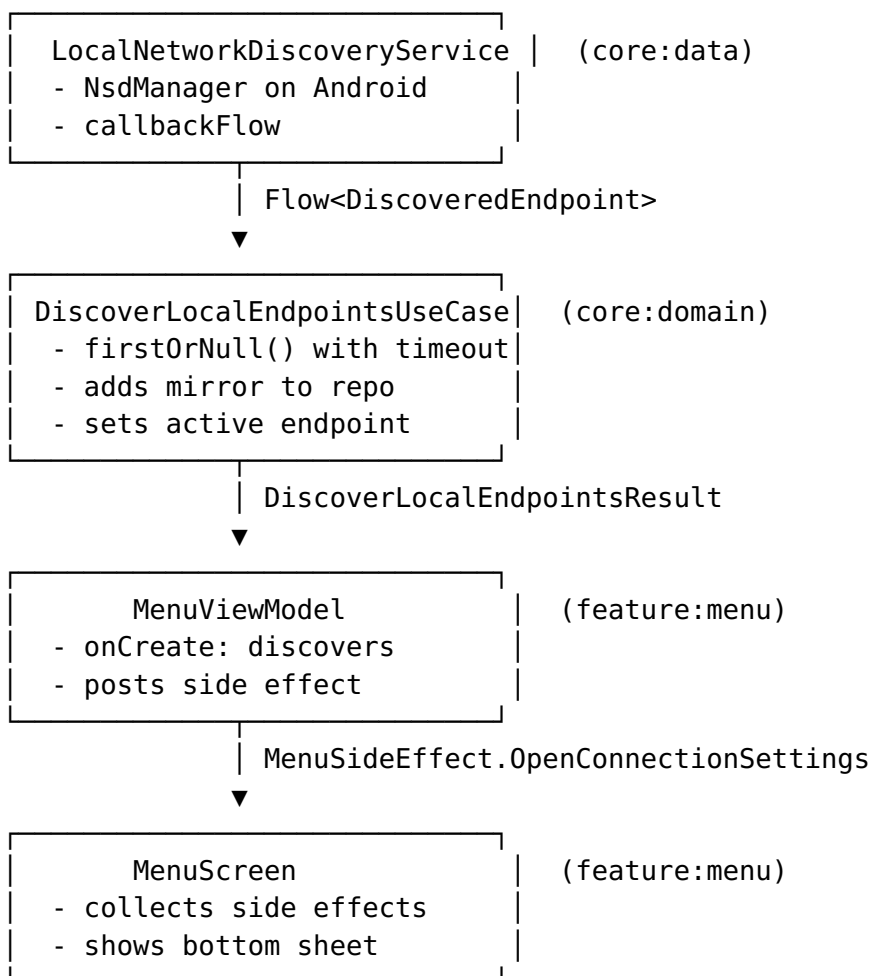
Local Network Discovery Architecture

Sequence Diagram

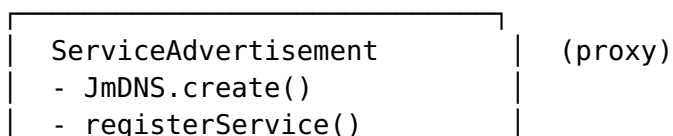




Data Flow



Proxy Advertisement



```
| - type: _lava._tcp.local. |
```

Module Dependencies

```
feature:menu
├── core:domain
│   ├── core:data
│   │   └── LocalNetworkDiscoveryService (NsdManager)
│   └── core:models
proxy
└── JmDNS advertisement
```

On-Device Lava API (Android) — third app + SQLite backend

A third artifact is being added alongside :app (the client) and the host lava-api-go server: a standalone **Lava API Android app** (:api-app, applicationId digital.vasic.lava.api, debug suffix .dev) that boots the full Lava API **in-process** on a phone/tablet, backed by a local SQLite database, and exposes it on the LAN via mDNS.

Landed (Phases A + B):

- **Additive SQLite storage backend** in lava-api-go. A storage.Storage interface (Get/Set/Invalidate/Close) with two interchangeable implementations — the existing Postgres adapter (unchanged) and a new pure-Go modernc.org/sqlite impl (WAL, INCREMENTAL auto_vacuum, 10-min background GC, lazy TTL expiry). Selected at startup by LAVA_API_STORAGE_BACKEND (postgres default | sqlite). The Postgres default behaves byte-for-byte as before.
- **In-process Go embed** (internal/mobile): Start/Stop/Status boot the EXACT production Gin router (shared internal/router.Build) over HTTP/1.1+H2 TLS, bound to 0.0.0.0, SQLite-backed, enforcing the identical Lava-Auth gate the host binary uses. Cross-compiled for Android via go build -buildmode=c-shared (gomobile is blocked by the relative replace ../submodules/* directives) plus a hand-written JNI bridge exposing digital.vasic.lava.apigo.LavaNative (nativeStart/Stop/Status).

Pending (Phases C/D/E): the :core:apiengine Kotlin wrapper, the :api-app Compose module, the foreground ApiEngineService, the on-device NsdManager advertiser (TXT engine=go, platform=android, storage=sqlite), the landing UI, and the instrumented Challenge tests.

Full detail and Mermaid diagrams (component, lifecycle, mDNS+auth sequence, build pipeline): [docs/ON_DEVICE_API.md](#).
User-facing guide: [docs/guides/ON_DEVICE_API_USER_GUIDE.md](#).